AE 370 Group Project 1

Numerical study of a perturbed Earth orbit

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1 Premise and equations

Satellites orbiting Earth are affected by atmospheric drag, Earth's nonspherical gravitational field, and the Moon's gravity. These effects can change a satellite's trajectory and require corrective maneuvers to maintain its orbit. We numerically study these changes and estimate the required velocity corrections.

We address the following questions with matched simulations at two altitudes:

1. How does atmospheric drag affect the satellite's orbital trajectory over time?
2. How does Earth's gravitational field, including the J_2 perturbation, affect the satellite's orbital trajectory?
3. How does the Moon's gravitational influence change the satellite's trajectory, and how does it compare with drag and J_2 at different altitudes?
4. How do the numerical method and time step affect the predicted trajectory, particularly when lunar gravity is included?
5. How much additional corrective thrust is required to maintain the desired orbit when the Moon's gravity is included?

The full model is a nonlinear, time-dependent ODE with no known closed-form solution. Drag depends on altitude and speed, and the Moon's position changes with time. We therefore solve an initial value problem numerically. A simpler circular orbit with only central gravity has an exact solution, which we use to check the implementation.

Let $\boldsymbol{r} = (x, y)^T$, $\boldsymbol{v} = (v_x, v_y)^T$, $r = |\boldsymbol{r}|$, and $h = r - R_E$. In an Earth-centered inertial frame, the state is

$$\boldsymbol{u} = (x, y, v_x, v_y)^T, \qquad \dot{\boldsymbol{u}} = f(t, \boldsymbol{u}) = \begin{bmatrix} \boldsymbol{v} \\ \boldsymbol{a}_g + \boldsymbol{a}_d + \boldsymbol{a}_{J_2} + \boldsymbol{a}_M \end{bmatrix}.$$

The orbit is equatorial, with $z = 0$. Central gravity and the equatorial J_2 correction are

$$\boldsymbol{a}_g = -\frac{\mu_E}{r^3}\boldsymbol{r}, \qquad \boldsymbol{a}_{J_2} = -\frac{3J_2\mu_E R_E^2}{2r^5}\boldsymbol{r}.$$

We assume a stationary atmosphere, so its relative velocity is $\boldsymbol{v}$:

$$\rho(h) = \rho_0 e^{-(h-h_0)/H}, \qquad \boldsymbol{a}_d = -\frac{C_D A \rho(h)}{2m}|\boldsymbol{v}|\boldsymbol{v}.$$

The prescribed lunar orbit and third-body acceleration are

$$\boldsymbol{r}_M(t) = D\begin{bmatrix}\cos(\omega_M t) \\ \sin(\omega_M t)\end{bmatrix}, \qquad \omega_M = \frac{2\pi}{T_M},$$

$$\boldsymbol{a}_M = \mu_M\left[\frac{\boldsymbol{r}_M - \boldsymbol{r}}{|\boldsymbol{r}_M - \boldsymbol{r}|^3} - \frac{\boldsymbol{r}_M}{D^3}\right].$$

The second lunar term subtracts the Moon's acceleration of Earth's center. Omitting it would give the wrong Earth-centered equations. The lunar phase starts at zero. We neglect the Sun, atmospheric rotation, and changes in satellite mass. Maneuvers are modeled as velocity jumps between ODE integration intervals, as described in Section 6.

2 Parameters and numerical method

All calculations use meters, seconds, and kilograms. The following are fixed model inputs, not fitted measurements. The density law is a simplified assumption rather than a detailed atmosphere model.

Parameter	Value	Parameter	Value
μ_E	$3.986004418 \times 10^{14}$ m^3/s^2	R_E	6.378137×10^6 m
J_2	1.08263×10^{-3}	μ_M	4.9028×10^{12} m^3/s^2
D	3.84400×10^8 m	T_M	27.32 days
m	100 kg	A, C_D	1 m^2, 2.2
ρ_0	4×10^{-12} kg/m^3	h_0, H	400 km, 60 km

We use initial altitudes of 300 km (low) and 35,786 km (high). Each uncontrolled comparison starts with $\boldsymbol{u}_0 = (r_0, 0, 0, \sqrt{\mu_E/r_0})^T$, where $r_0 = R_E + h_{\text{initial}}$. This is circular for central gravity alone. It is deliberately the same initial state in every physical model, so it need not be circular after J_2 is added.

From Lecture 10, integrate the ODE across a step:

$$\boldsymbol{u}(t_{k+1}) - \boldsymbol{u}(t_k) = \int_{t_k}^{t_{k+1}} f(t, \boldsymbol{u}(t))\, dt.$$

Approximating this integral with staged slopes gives the classical RK4 method:

$$\begin{aligned} k_1 &= f(t_k, \boldsymbol{u}_k), & k_2 &= f(t_k + \Delta t/2, \boldsymbol{u}_k + \Delta t k_1/2), \\ k_3 &= f(t_k + \Delta t/2, \boldsymbol{u}_k + \Delta t k_2/2), & k_4 &= f(t_k + \Delta t, \boldsymbol{u}_k + \Delta t k_3), \\ \boldsymbol{u}_{k+1} &= \boldsymbol{u}_k + \frac{\Delta t}{6}(k_1 + 2k_2 + 2k_3 + k_4). \end{aligned}$$

Each step evaluates these four slopes in order and then updates the state. The time-dependent Moon position is recomputed at each stage time. The integrator is written from scratch, with no outside ODE solver.

For comparison, Heun uses $k_1 = f(t_k, \boldsymbol{u}_k)$ and a forward Euler prediction $\boldsymbol{u}^* = \boldsymbol{u}_k + \Delta t k_1$, followed by

$$\boldsymbol{u}_{k+1} = \boldsymbol{u}_k + \frac{\Delta t}{2}[k_1 + f(t_k + \Delta t, \boldsymbol{u}^*)].$$

It costs two evaluations per step, compared with four for RK4. Taylor expansion gives $f(t + \Delta t, u + \Delta t f) = f + \Delta t(f_t + f_u f) + O(\Delta t^2)$. Thus Heun matches $u(t+\Delta t) = u + \Delta t f + \Delta t^2(f_t + f_u f)/2 + O(\Delta t^3)$: its one-step error is $O(\Delta t^3)$ and global order is two for a smooth problem over a fixed interval. RK4 has one-step error $O(\Delta t^5)$ and global order four, as covered in Lecture 10.

Following Lecture 11, truncation error is the residual after substituting the exact solution, divided by Δt. For RK4 this is $O(\Delta t^4)$, rather than the undivided $O(\Delta t^5)$ step error. The numerical checks below test the expected global orders on the nonlinear orbital equations.

3 Stability and independent validation

For the lecture test equation $\dot{u} = \lambda u$, let $w = \Delta t\lambda$. Substitution into RK4 gives successive slopes proportional to

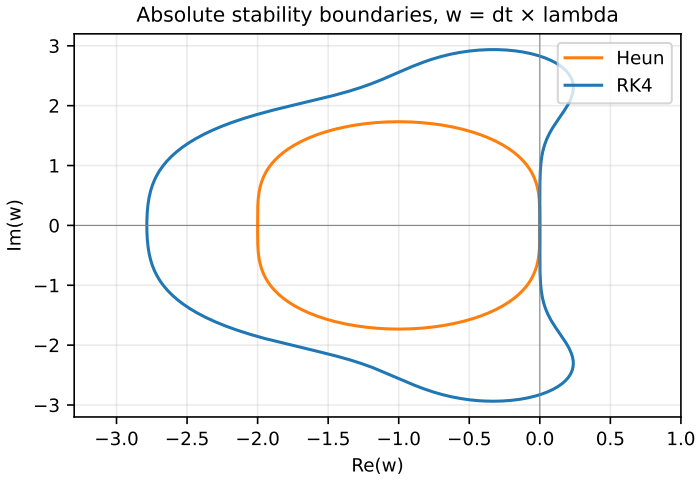
$$\lambda u_k, \quad \lambda u_k(1 + w/2), \quad \lambda u_k(1 + w/2 + w^2/4), \quad \lambda u_k(1 + w + w^2/2 + w^3/4).$$

Collecting terms gives $u_{k+1} = R(w)u_k$, where

$$R_{\mathrm{RK4}}(w) = 1 + w + \frac{w^2}{2} + \frac{w^3}{6} + \frac{w^4}{24}, \quad R_{\mathrm{Heun}}(w) = 1 + w + \frac{w^2}{2}.$$

RK4 matches e^w through degree four. Its test-equation truncation error is $\tau_k = \lambda^5 u(t_k)\Delta t^4/120 + O(\Delta t^5)$. This verifies the order on the linear test problem; it alone is not a proof for every nonlinear ODE.

Absolute stability requires $|R(w)| < 1$ for decay. On the imaginary axis, writing $w = iq$ gives $|R_{\mathrm{RK4}}(iq)|^2 = 1 - q^6/72 + q^8/576$. Thus RK4 has no amplification for $|q| \le \sqrt{8}$. Heun instead gives $1 + q^4/4 > 1$ for nonzero q, which can cause growth in oscillatory calculations. Stability does not guarantee small phase or position error.



The curves bound the regions containing small negative real w. Our orbit is nonlinear and time dependent, so a fixed scalar test does not establish its full stability. We do not claim stiffness simply because the Moon and satellite have different periods. RK4 is a practical explicit choice, with reliability checked by refinement.

For independent validation, remove drag, J_2, and the Moon. At 300 km the circular solution is

$$\boldsymbol{r}(t) = r_0(\cos nt, \sin nt)^T, \qquad n = \sqrt{\mu_E/r_0^3}, \qquad P = 2\pi/n = 5431.18 \text{ s}.$$

We compare over three periods using $E = \max_k |\boldsymbol{r}_k - \boldsymbol{r}(t_k)|$. The known solution checks correctness as well as convergence. We also check the specific energy $\epsilon = |\boldsymbol{v}|^2/2 - \mu_E/r$ and specific angular momentum $\ell = xv_y - yv_x$, which must be constant in this special case. At the finest RK4 step, their maximum relative changes are 1.49×10^{-12} and 7.45×10^{-13}.

4 Convergence and choice of time step

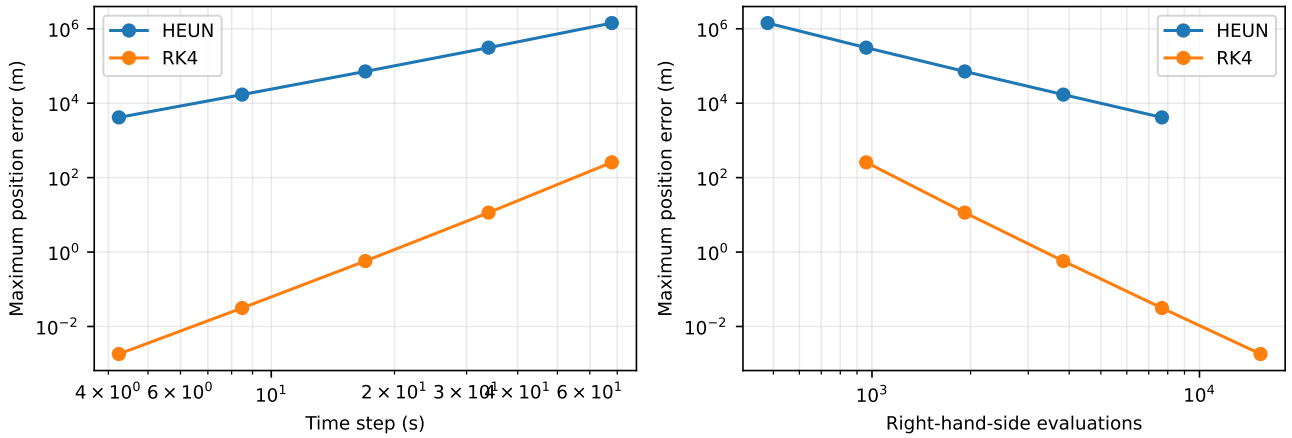


Figure 1. Exact circular-orbit position errors over three periods. The right panel compares cost by the number of ODE right-hand-side evaluations.

Δt (s)	Heun error (m)	RK4 error (m)
67.8897	1.4230×10^6	257.367
33.9449	3.1083×10^5	11.4748
16.9724	7.1348×10^4	0.57301
8.4862	1.7016×10^4	0.031311
4.2431	4.1529×10^3	0.0018218

Using $p = \log_2(E(\Delta t)/E(\Delta t/2))$ on the last pair gives $p = 2.03$ for Heun and $p = 4.10$ for RK4. The rates approach the expected two and four. At equal cost of 7680 evaluations, Heun's error is 4153 m while RK4's is 0.0313 m. The measured runtimes of these two short runs were about 0.015 s and 0.017 s; such short timings are machine dependent. Evaluation counts are the clearer comparison.

For seven-day results we select $\Delta t = 5$ s at low altitude and 120 s at high altitude. The full model and the baseline are each repeated at half that step. The maximum coarse–fine differences at common times are:

Orbit	Position, full model (m)	Altitude, full model (m)	Paired lunar displacement difference (m)
Low	1.3773	0.004052	0.000222
High	0.3185	0.009618	0.000438

The last column compares $(\boldsymbol{r}_M - \boldsymbol{r}_B)$ between the two step sizes, where the subscripts here mean the full lunar run and baseline run. It checks the actual difference we interpret as the Moon's effect. These are numerical differences, not rigorous error bounds. Both the individual-run and paired differences are much smaller than the lunar signals reported next. For scale, $n\Delta t$ is about 0.0058 in the low orbit and 0.0087 in the high orbit. These are small relative to the imaginary-axis test limit, but the refinement check is still necessary.

5 Physical results over seven days

We compare central gravity alone, central gravity with drag, central gravity with J_2, and the baseline containing all three. A fifth run adds the Moon to the baseline. No corrections are applied in this section.

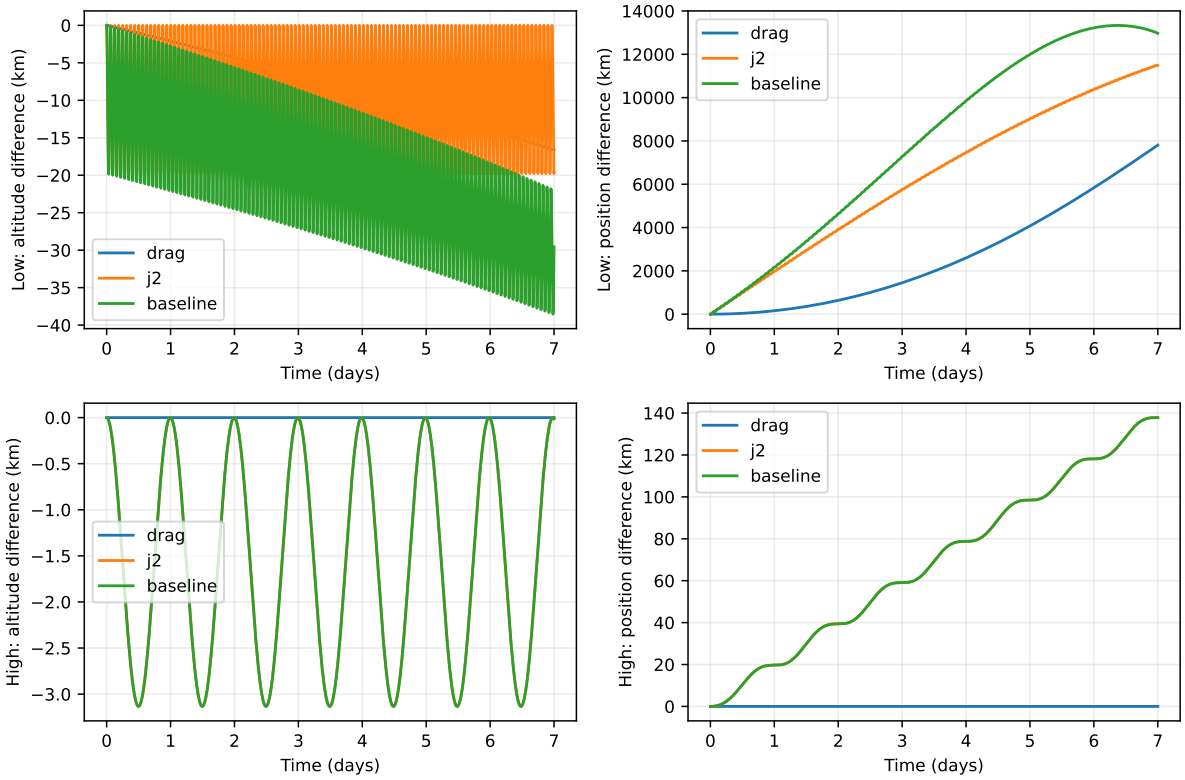


Figure 2. Differences from central gravity alone. The left panels show signed altitude differences; the right panels show position-separation magnitudes. Curves can overlap.

With drag alone, the low orbit loses 16.643 km of altitude by day seven. Drag opposes velocity: $\boldsymbol{v} \cdot \boldsymbol{a}_d = -C_D A \rho |\boldsymbol{v}|^3/(2m) \leq 0$, so it removes orbital energy. The high-orbit density is effectively zero in this exponential model and its drag-only run agrees with central gravity to numerical precision. Extending one density law over this full altitude range is a modeling simplification.

Adding J_2 alone gives bounded radial oscillations, reaching 19.785 km below the initial low altitude and 3.134 km below the initial high altitude. The initial central-gravity circular speed is too small for a circular orbit in the stronger equatorial gravity field. These oscillations therefore do not mean that J_2 dissipates energy. The central-gravity energy formula is not the full conserved energy after adding J_2. The altered orbital phase also produces large position separations. Our planar model does not describe changes in the orbital plane.

With drag and J_2 together, the low altitude reaches a maximum departure of 38.542 km from its initial value, ending at 270.427 km. The combined behavior is not a simple sum: lower altitude increases the assumed density and hence the drag. At high altitude the baseline coincides with the J_2-only result.

Adding lunar gravity

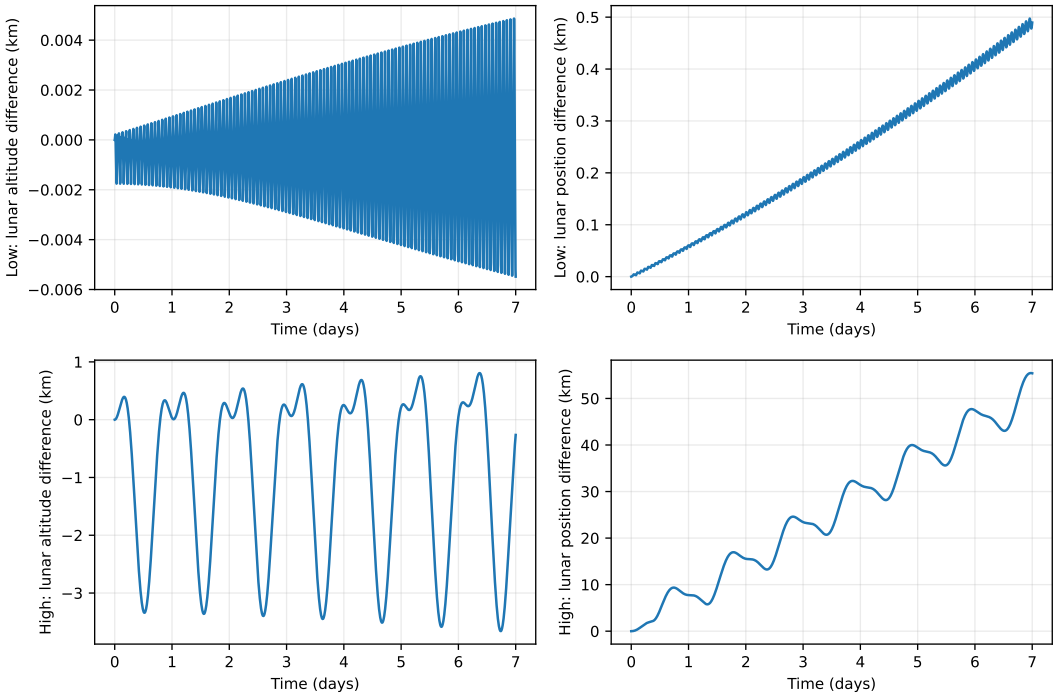


Figure 3. Full lunar model minus the baseline, with identical initial conditions. Position separation includes phase differences as well as radial differences.

Seven-day maximum	Low orbit	High orbit
Lunar position separation (km)	0.4974	55.457
Absolute lunar altitude difference (km)	0.005484	3.65995

The Moon changes the low-orbit altitude by at most about 5.5 m relative to the baseline, much less than the drag and J_2 effects. Its position difference reaches about 497 m because small changes in motion can accumulate into phase differences. At high altitude the lunar altitude change is comparable to the J_2-only radial excursion, while atmospheric drag is negligible in this model. The lunar position separation reaches about 55.5 km.

This comparison uses the differential lunar acceleration, not the Moon's direct pull on the satellite alone. Both Earth and satellite respond to the Moon, and that common acceleration is removed. Over larger Earth-centered distances their lunar accelerations differ more strongly. This is consistent with the greater lunar effect in our high orbit.

We also ran Heun on the full lunar model, comparing it with half-step RK4. At low altitude, its maximum position difference falls from 528.5 km at 5 s to 98.19 km at 2.5 s. At high altitude it falls from 209.9 km at 120 s to 49.91 km at 60 s. These errors exceed the lunar position signal. A smooth-looking trajectory at these steps would therefore be insufficient evidence of accuracy; method choice matters even when the run does not fail.

The half-step differences in Section 4 are below these signals. However, that only addresses integration error. It does not remove uncertainty caused by the circular Moon orbit, chosen initial phase, or simplified atmosphere. The results apply to this seven-day experiment rather than every satellite or lunar geometry.

6 A simple orbit-maintenance experiment

Every 600 s we reset velocity while leaving position continuous. Let r_0 be the initial radius, $\boldsymbol{e}_r = \boldsymbol{r}/r$, and $\boldsymbol{e}_t = (-y, x)^T/r$. We prescribe

$$v_c(r) = \sqrt{\frac{\mu_E}{r}\left[1 + \frac{3}{2}J_2\left(\frac{R_E}{r}\right)^2\right]}, \qquad \boldsymbol{v}^+ = v_c(r)\boldsymbol{e}_t - \frac{r - r_0}{3600\ \mathrm{s}}\boldsymbol{e}_r.$$

The circular speed follows by setting v_c^2/r equal to the magnitude of central-plus-J_2 radial gravity. The radial term points back toward the initial altitude. The checking interval and return time are fixed assumptions; no optimization or advanced control method is used. At each correction,

$$\Delta v_j = |\boldsymbol{v}^+ - \boldsymbol{v}^-|, \qquad \Delta v_{\mathrm{total}} = \sum_j \Delta v_j.$$

We use the same rule in the baseline and lunar runs. There are 1008 corrections over seven days, including the correction at the final time. The rule maintains altitude, not a specified longitude or orbital phase. We use a 3 km altitude band as a simple success criterion, checking the sampled trajectory between maneuvers as well as at them.

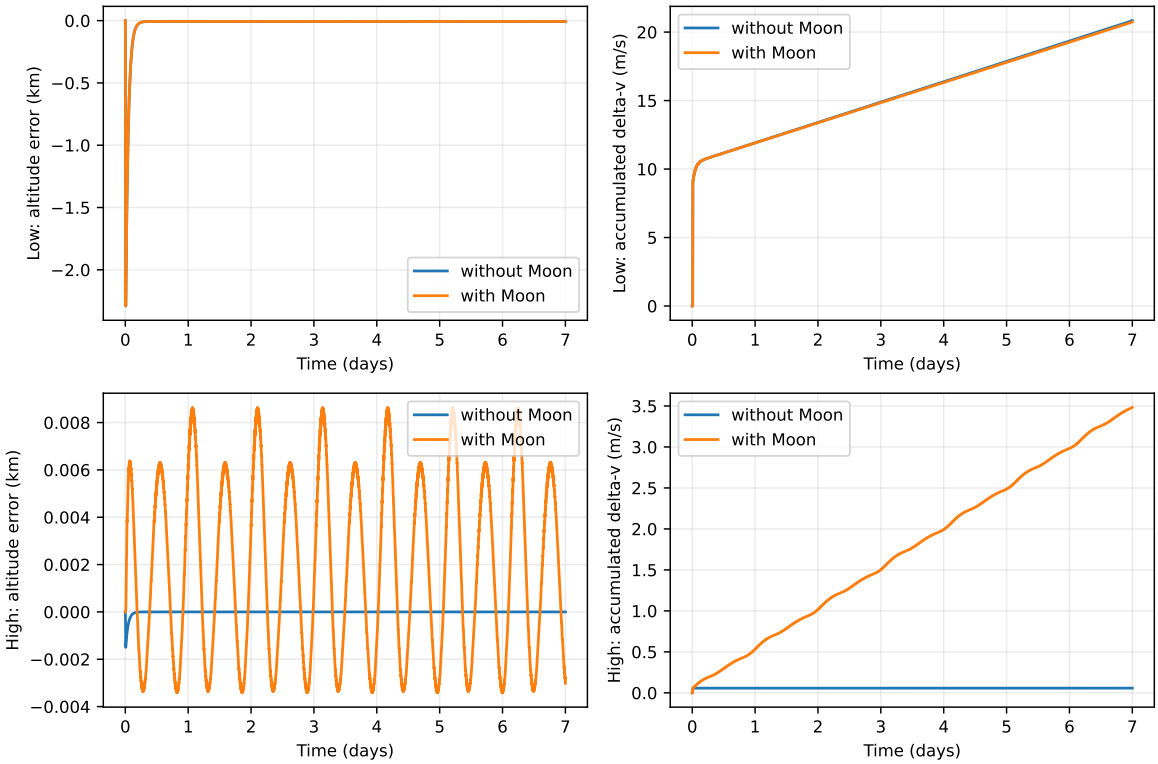


Figure 4. Altitude errors and accumulated maneuver magnitudes. The nearly overlapping low-orbit curves are expected because its lunar effect is small.

Quantity	Low orbit	High orbit
Baseline total Δv (m/s)	20.8366	0.05776
Lunar total Δv (m/s)	20.7389	3.48176
Lunar minus baseline Δv (m/s)	−0.09760	+3.42400
Maximum lunar-run altitude error (km)	2.29046	0.008638

7 Answers to the project questions

The answers below follow the numbering in the premise.

1. **Atmospheric drag.** Drag removes orbital energy and lowers the low orbit. With drag alone, its altitude falls by 16.643 km in seven days. Drag is negligible at the high altitude under the assumed density law (Section 5, Figure 2).

2. **Earth's gravity and J_2.** Central gravity alone maintains the initial circular orbit. Adding J_2 strengthens equatorial gravity, changes the orbital phase, and produces radial oscillations because the same initial speed is no longer circular. Maximum altitude departures with J_2 alone are 19.785 km in the low orbit and 3.134 km in the high orbit. This is not drag-like energy loss (Section 5, Figure 2).

3. **Lunar gravity.** Relative to the baseline, the Moon produces maximum position separations of 0.4974 km and 55.457 km in the low and high orbits. Maximum altitude differences are 5.484 m and 3.65995 km. Thus the lunar radial effect is much smaller than drag and J_2 at low altitude, but comparable to the J_2 radial excursion at high altitude, where drag is negligible (Section 5, Figure 3).

4. **Method and time step.** Independent validation gives convergence orders of 2.03 for Heun and 4.10 for RK4. With the Moon included, halving Heun's step reduces its maximum position difference from the fine RK4 reference from 528.5 to 98.19 km in the low orbit and from 209.9 to 49.91 km in the high orbit. The selected RK4 steps give coarse–fine position differences of only 1.3773 m and 0.3185 m. Method choice and refinement are therefore necessary to distinguish lunar effects from numerical error (Sections 4–5).

5. **Corrective maneuvers.** For the same maintenance rule, adding the Moon changes the seven-day total Δv by -0.09760 m/s in the low orbit and $+3.42400$ m/s in the high orbit. Both controlled lunar runs stay within the selected 3 km altitude band at sampled times (Section 6, Figure 4). These are impulse costs, not engine forces: force would also require specifying burn durations. The negative low-orbit difference means that the Moon partly offsets other deviations for this phase and rule; it is not a general reduction in maintenance cost.

Limits of these answers

Halving the step changes the signed maneuver-cost differences by 1.15×10^{-6} m/s at low altitude and 1.23×10^{-5} m/s at high altitude, well below the reported changes. These checks address integration error, not physical-model uncertainty. The results use one lunar phase, a prescribed circular Moon orbit, a simplified atmosphere, and fixed satellite parameters. The planar model cannot describe orbital-plane changes. The impulse totals include adjustment from the initial central-gravity circular orbit to the J_2 model and are not optimal station-keeping costs.

8 Reproducibility and contributions

The submission contains `simulation.py`, `results.json`, the figures, this LaTeX source, and the compiled PDF. Running `python simulation.py` reproduces all calculations and figures. The integrators, checks, experiments, and plotting are separate functions within one script. Only NumPy and Matplotlib are required. To rebuild the PDF after generating the figures, compile `report.tex` with LaTeX.

The group members are Jiho Seo, Pranshu Bhardwaj, Adi Jha, and Dmitry Moryakov.

Generative AI

OpenAI ChatGPT assisted with interpreting the supplied materials, planning the experiments, implementing and checking the code, producing figures, and drafting and compiling this report.